Suchakra Kumar Gattu

AI Engineer: Voice AI, LLM Inference and LLMOps

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SUMMARY

AI Engineer at CloudAngles leading **One08labs**, a self-hosted voice AI platform with approximately 1.15 s end-to-end latency that works with any STT, LLM, TTS, or telephony provider, and building LLMOps infrastructure on **mlangles** (fine-tuning, inference serving, quantization, RAG evaluation). Also replaced Agora with self-hosted LiveKit for a UK client, cutting cost by about 30% and scaling past a measured 1,000-user per-node limit with a binary-tree cluster design aimed at 1M users.

EDUCATION

Keshav Memorial Engineering College, Osmania University **Hyderabad**

B.E., Computer Science and Engineering (Specialization: AI & ML) *2021 – 2025*

- **Coursework:** Machine Learning, Deep Learning, Operating Systems, Computer Networks, DBMS, Data Structures and Algorithms, Information Retrieval

TECHNICAL SKILLS

Languages	Python, TypeScript, JavaScript, Java, C
Voice & Real-Time	LiveKit (Server, SIP, Ingress, Egress, Agents), WebRTC, SIP Trunking, STT/TTS Pipelines, Twilio, Vonage, Vobiz
LLM Inference	vLLM, SGLang, TensorRT, NVIDIA Dynamo, llama.cpp, ONNX, GGUF Quantization (Q4_K_M), Edge Inference Deployment
AI / ML	PyTorch, TensorFlow, Transformers, LoRA, QLoRA, PEFT, SFT, DPO, RAG, LangChain, LangGraph
MLOps	MLflow, Langfuse, Opik
Backend	FastAPI, NestJS, Node.js, Express.js, Flask, Django, SQLAlchemy, Alembic, Prisma
Infrastructure & Reliability	Linux (Ubuntu), Docker, AWS (EC2, ECS, EBS, Bedrock, Security Groups), Dedicated-Server Hosting, Prometheus, Grafana, cAdvisor, Node Exporter
Frontend & Databases	React, React Router DOM, TanStack Router, TanStack Query; PostgreSQL, MySQL, MongoDB

WORK EXPERIENCE

CloudAngles **Aug 2025 – Present**

Software Engineer *Hyderabad*

One08labs: Self-Hosted Voice AI Platform (Lead) | *LiveKit, WebRTC, SIP*

- Lead engineering and act as the primary technical point of contact (SPOC) for enterprise customers: gathering requirements, scoping solutions, and shipping frequent releases.
- Self-hosted the complete LiveKit stack (Server, SIP, Ingress, Egress) and built the voice-agent orchestration layer on it: a cascaded STT, LLM, and TTS pipeline over WebRTC, with SIP bridging so one platform serves both web and telephone calls.
- Reached approximately 1.15 s end-to-end voice latency on the self-hosted stack, with STT, LLM, and TTS providers fully swappable per customer.
- Cut voice platform cost by about **60%** versus Vapi by moving to the self-hosted LiveKit stack, currently serving 10 channels in production.
- Integrated CPaaS and SIP trunk providers (Twilio, Vonage, Vobiz, and others) with call routing for reliable inbound and outbound AI voice calls, avoiding lock-in to any single telephony vendor.

LudoStream-UK: Agora to Self-Hosted LiveKit | *LiveKit Server, Egress, WebRTC*

- Replaced Agora with a self-hosted LiveKit stack (LiveKit Server and Egress) for a UK client, cutting cost by about **30%**.
- Load-tested the recommended LiveKit machine spec, found a ceiling of about **800 to 1,000 users per room**, and removed it with a **binary-tree topology** of cluster nodes and load-based user distribution, designed to scale a single WebRTC session to **1M users**.
- Kept bandwidth cost sustainable at scale by running WebRTC media on dedicated machines at a separate cloud provider, where bandwidth is far cheaper than on AWS.

mlangles: LLMOps Platform | *Inference, Fine-Tuning, Quantization*

- Architected infrastructure for **mlangles**, CloudAngles' LLMOps platform: staging environments, model hub, and automated deployment pipelines, with dataset management, experiment tracking, and model versioning supporting internal and customer-facing use cases.
- Built a fine-tuning framework supporting SFT, DPO, and full fine-tuning, plus parameter-efficient methods (LoRA, QLoRA).
- Implemented self-hosted LLM inference serving on **vLLM, SGLang, and TensorRT**, and applied **NVIDIA Dynamo** to disaggregate prefill and decode across separate devices.
- Built quantization and runtime-conversion pipelines (llama.cpp, ONNX, TensorRT), including Q4_K_M GGUF conversion for on-device edge inference, shipped as a production platform feature.
- Implemented evaluation pipelines for RAG applications covering hallucination detection, answer relevance, and context scoring.
- Integrated Prometheus, Grafana, cAdvisor, and Node Exporter for production monitoring and observability.

CloudAngles **Mar 2025 – Jul 2025**

GenAI Developer Intern *Hyderabad*

- Built a centralized monitoring dashboard for internal AI products.
- Integrated customized computer vision models with Roboflow for dynamic image classification.
- Built LangGraph AI agents to automate enterprise workflows, and reusable UI components for chatbot and agent creation platforms.

Prathibha Innovations **Dec 2024 – Jan 2025**

SDE Intern *Hyderabad (Remote)*

- Built a custom user management system with Express.js, using Prisma ORM for scalable database operations.

AWARDS & HONORS

Gold Medal, State Level Science Talent Search Examination (SLSTE)